OpenAI

# Learning Dexterity

## OpenAI Robotics

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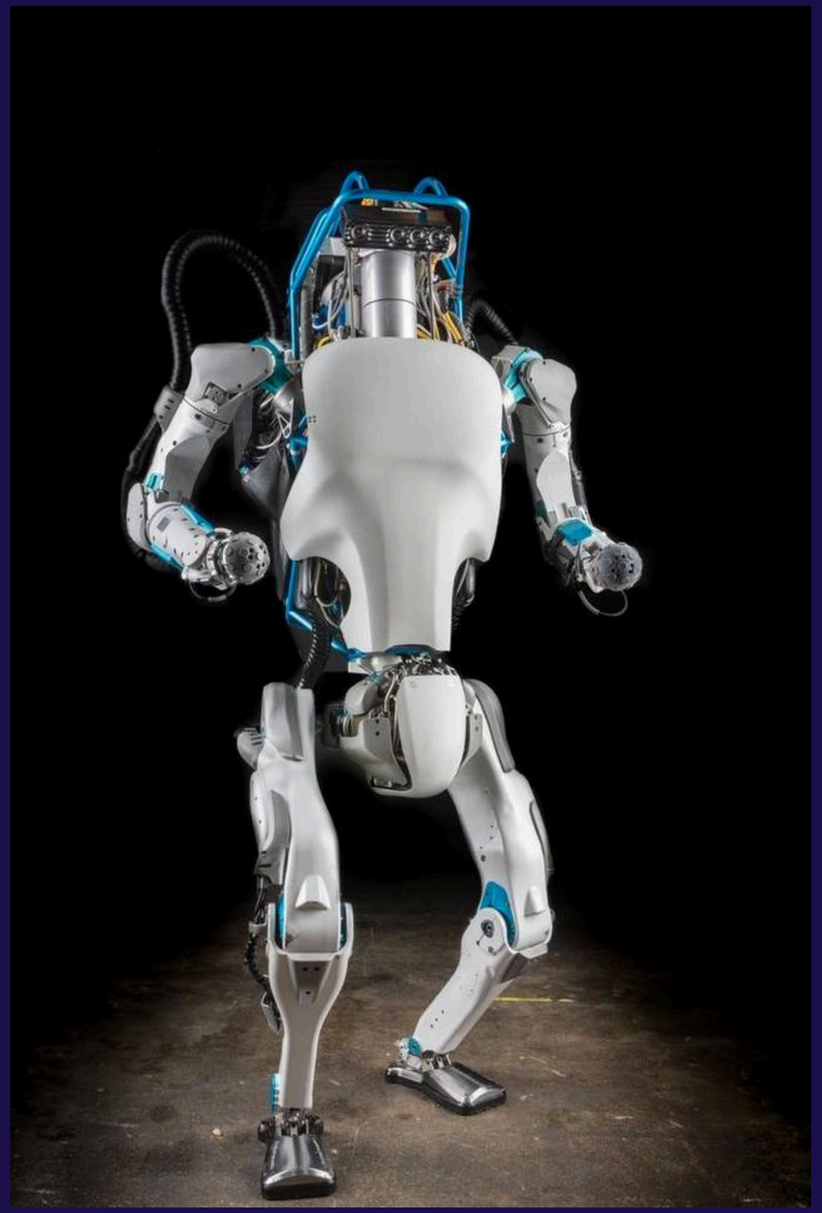
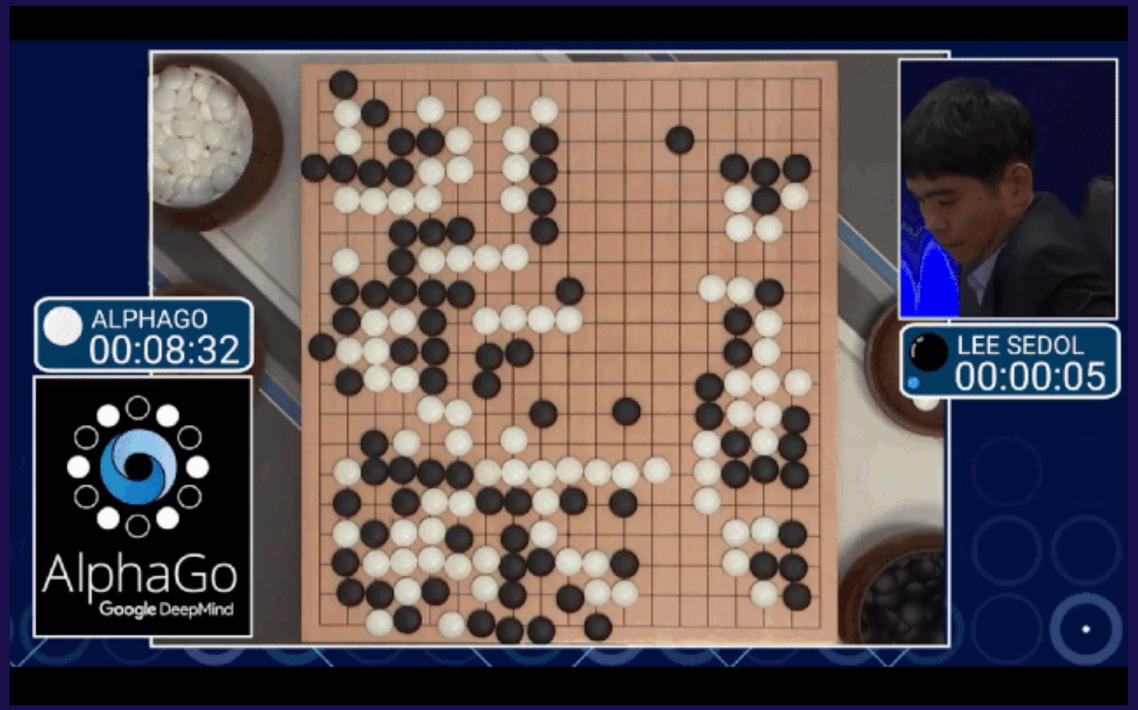
**SPINNING UP WORKSHOP, FEBRUARY 2, 2019**



FRAGILE
FRAGILE

WAYMO

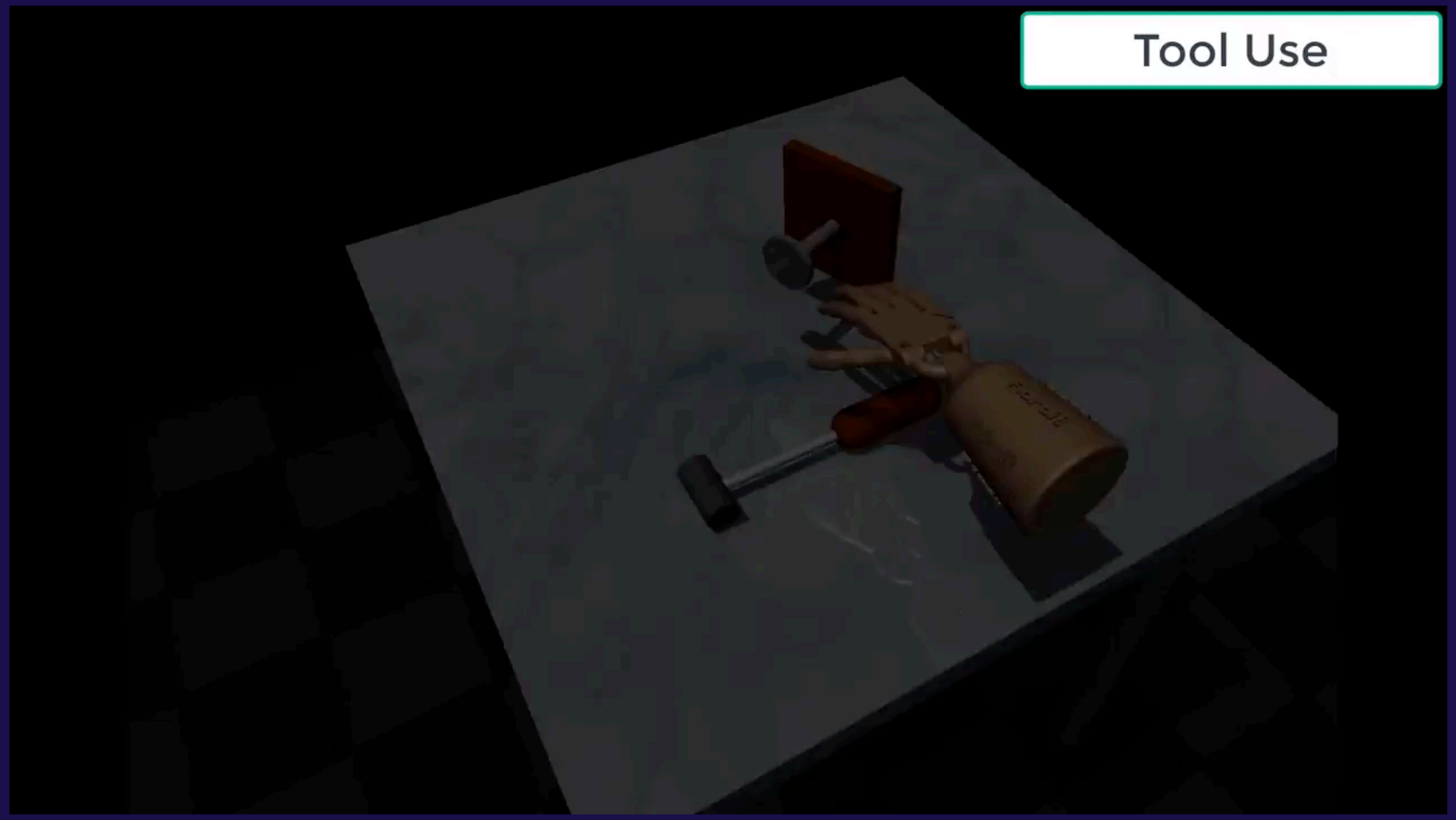
# Reinforcement Learning (RL)




GO (ALPHAGO ZERO)

DOTA 2 (OPENAI FIVE)

# RL for Robotics (1)




Rajeswaran et al. (2017)

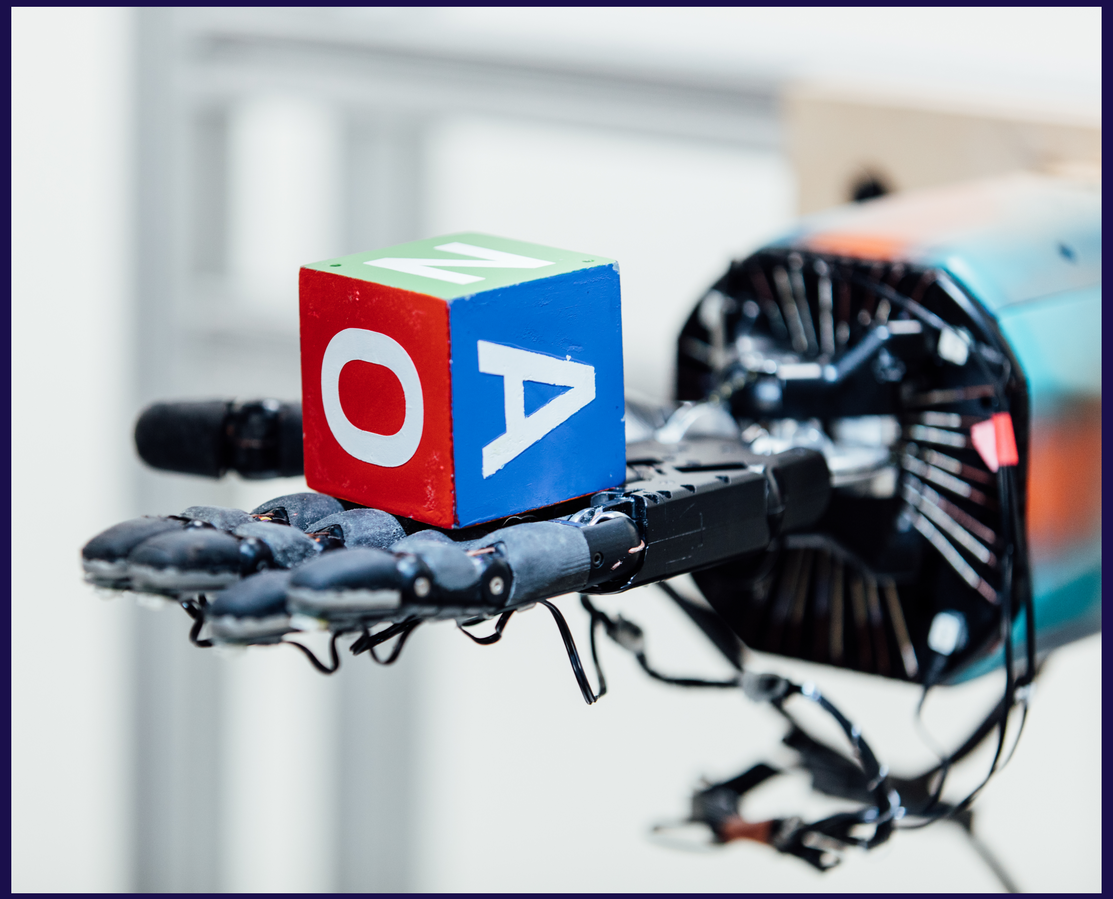
# RL for Robotics (2)

Kumar et al. (2016)

# RL for Robotics (3)

Levine et al. (2018)

# Dexterous In-Hand Manipulation



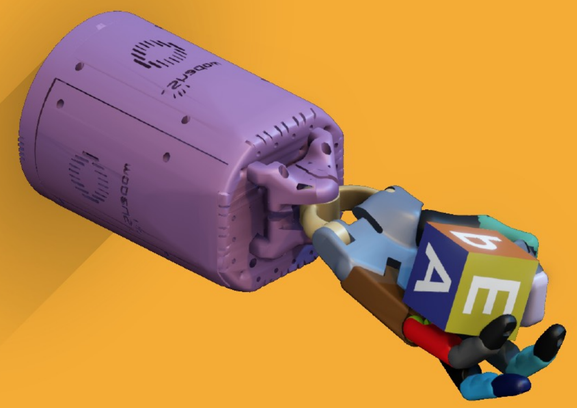

- A humanoid 5-fingered hand
- A human hand is a universal end-effector
- Long standing unachieved goal for classical robotics

# Dexterous In-Hand Manipulation

- High-dimensional control
- Real-world hardware
- Noisy and delayed sensor readings
- Partial observability
- Hard to simulate

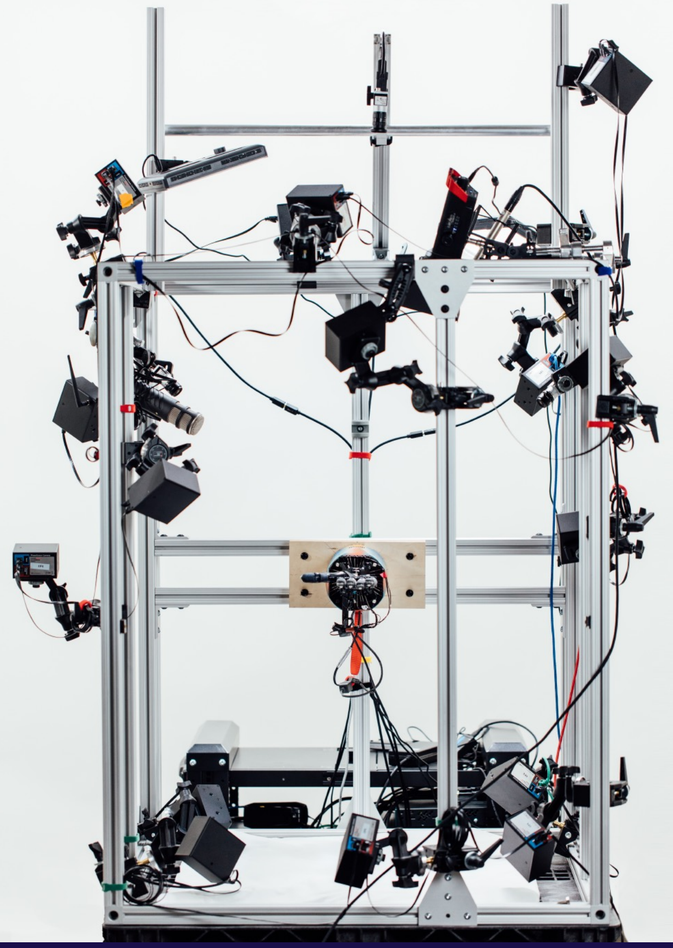
# Sim2Real

SIMULATION ENVIRONMENT

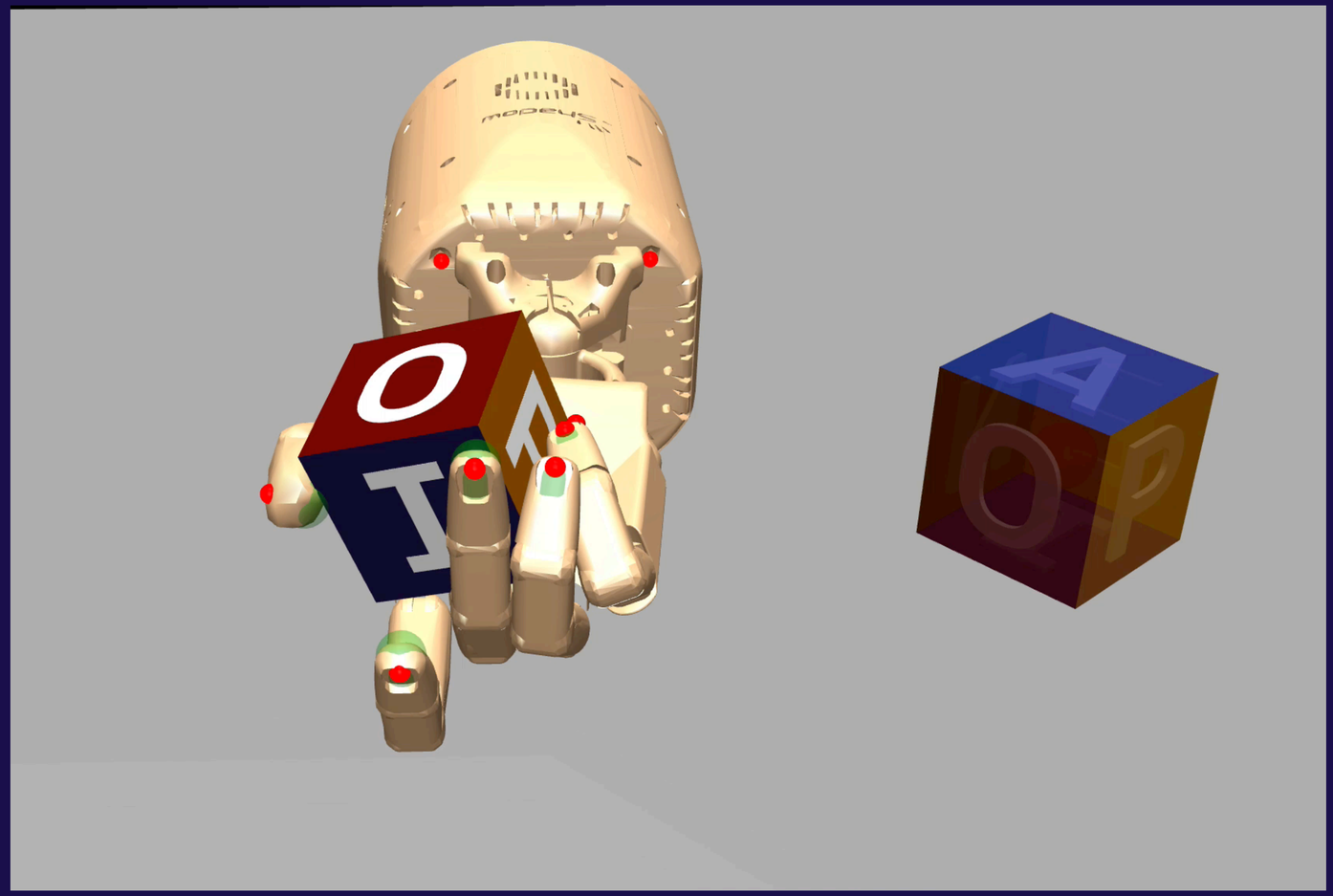


Transfer

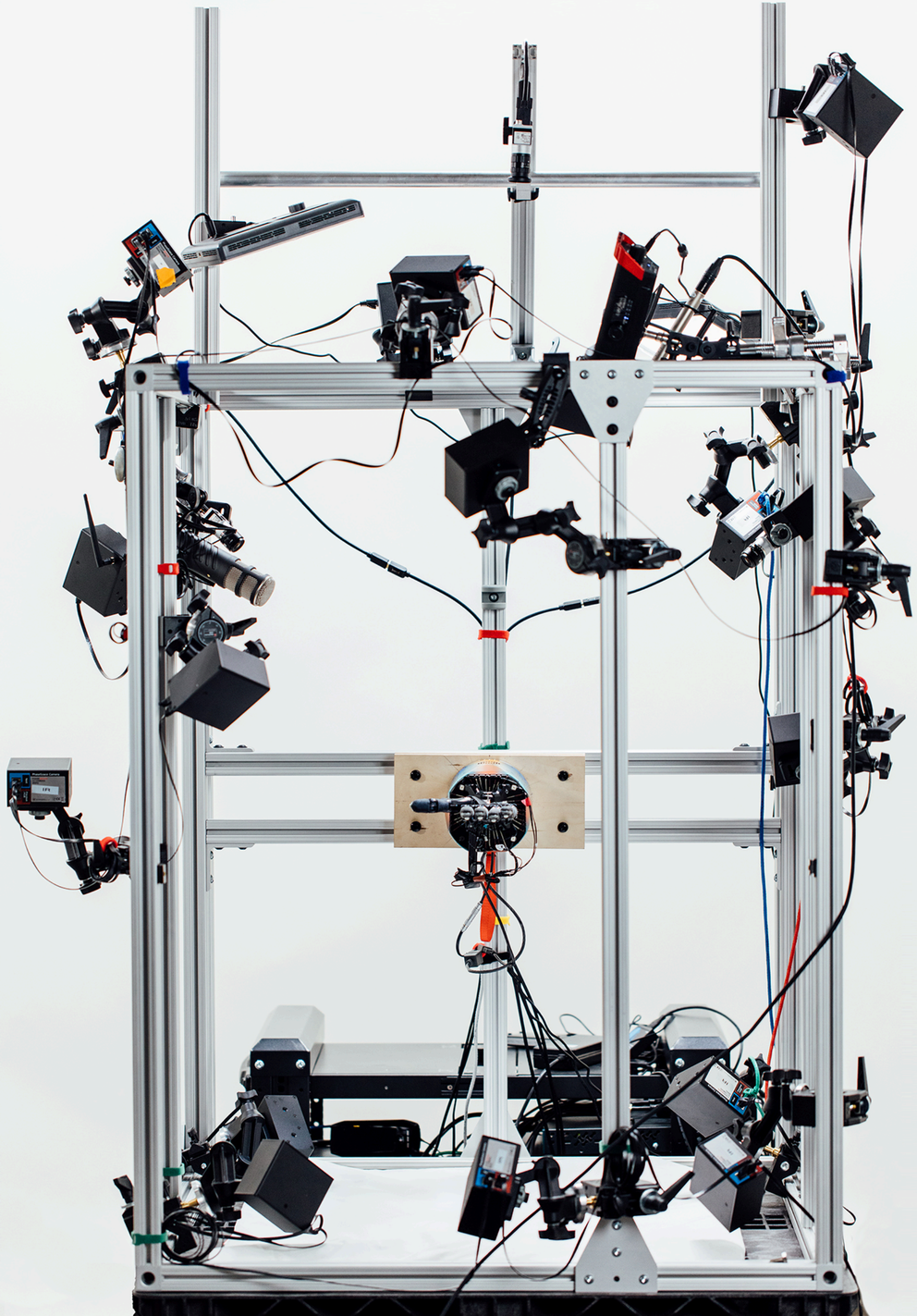
REAL-WORLD ENVIRONMENT

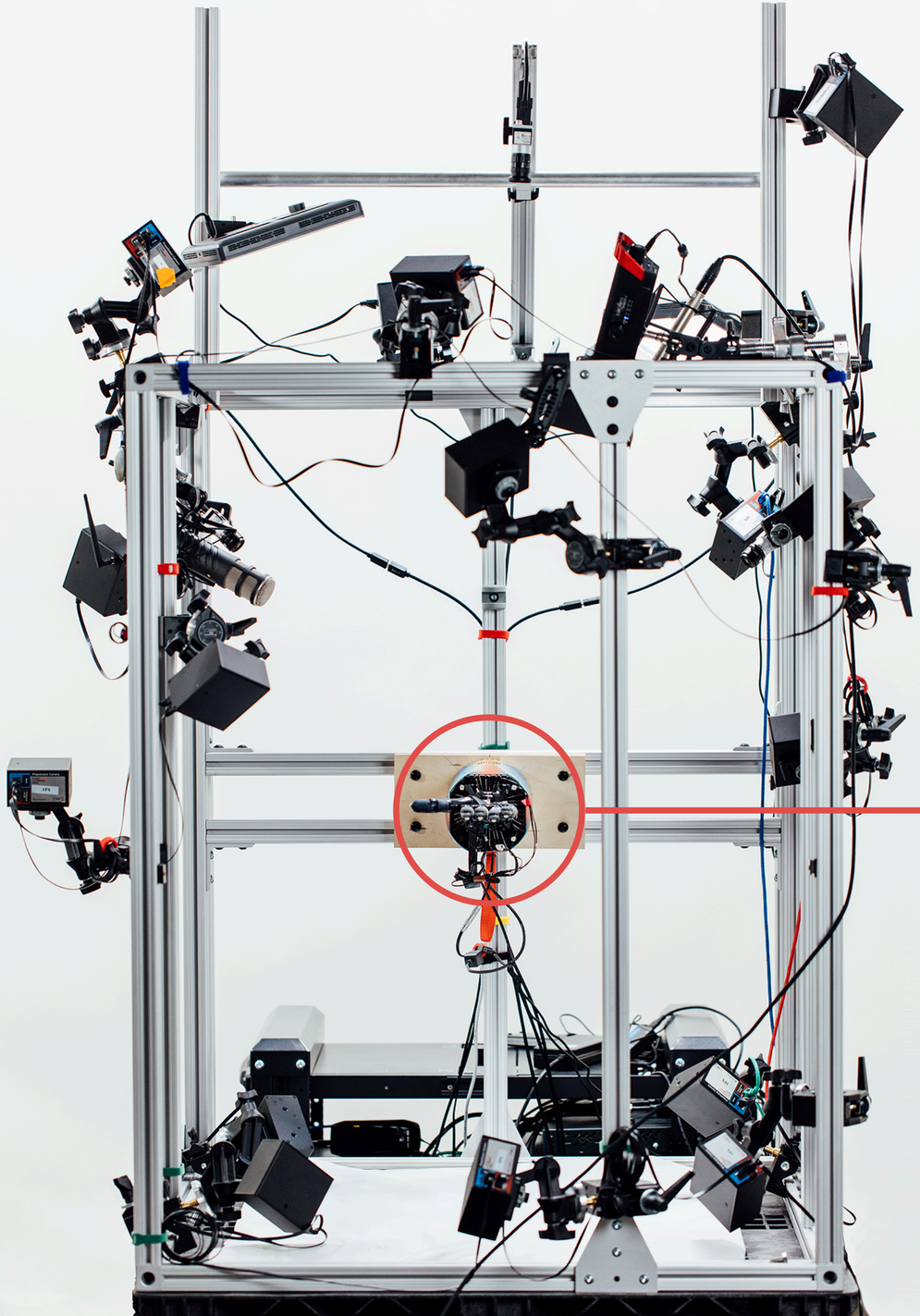


# Simulation



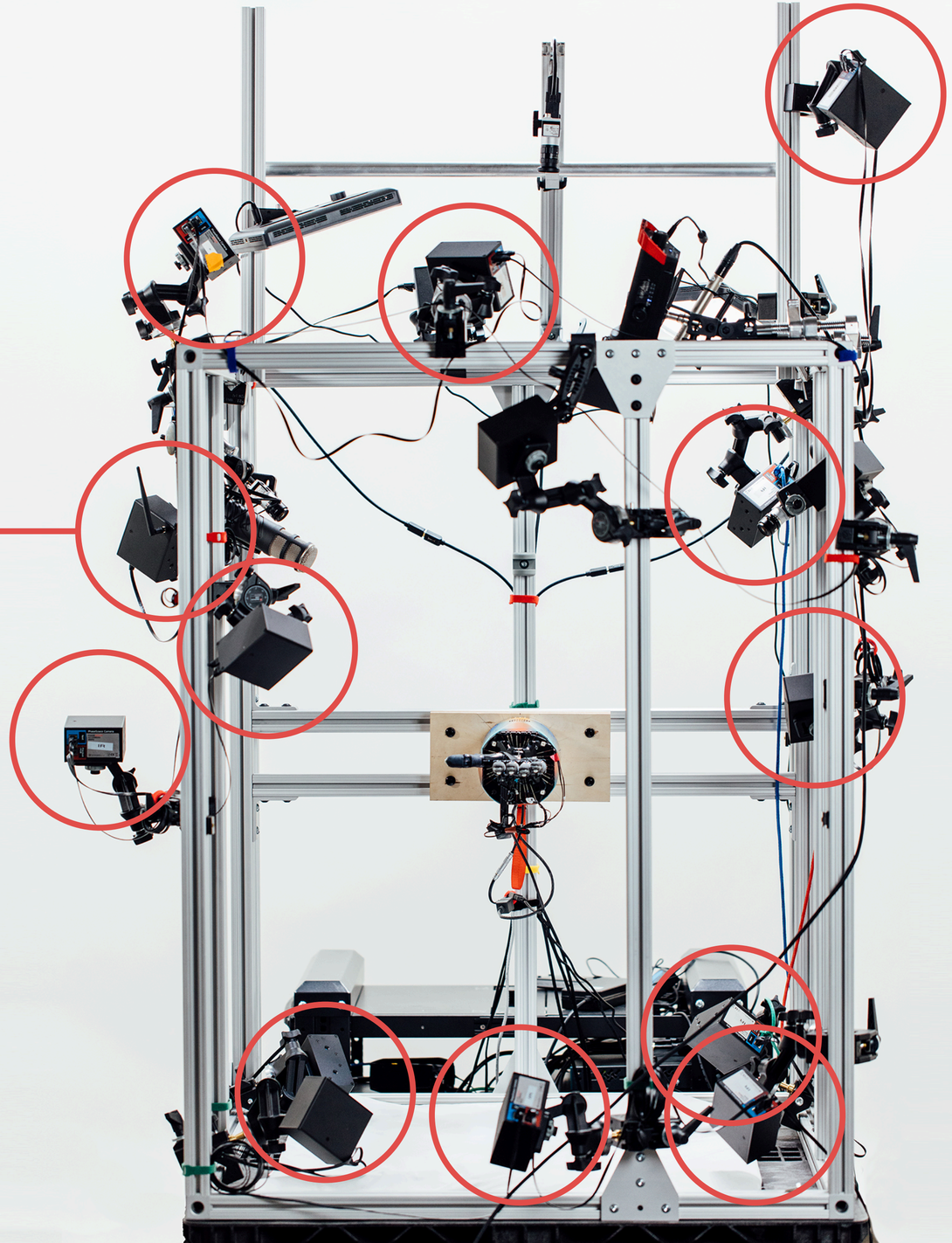
Task: reorient the object in-hand

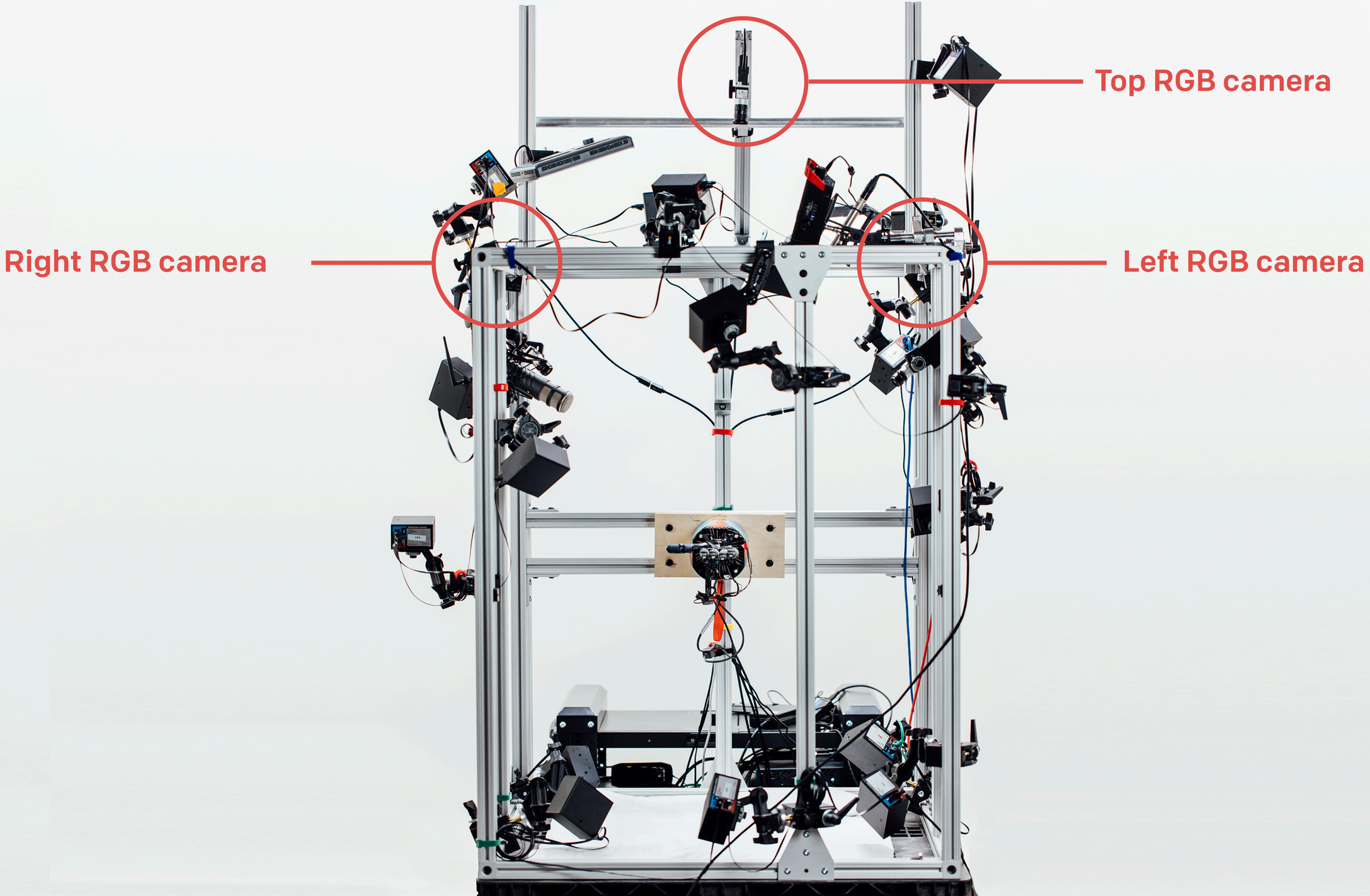


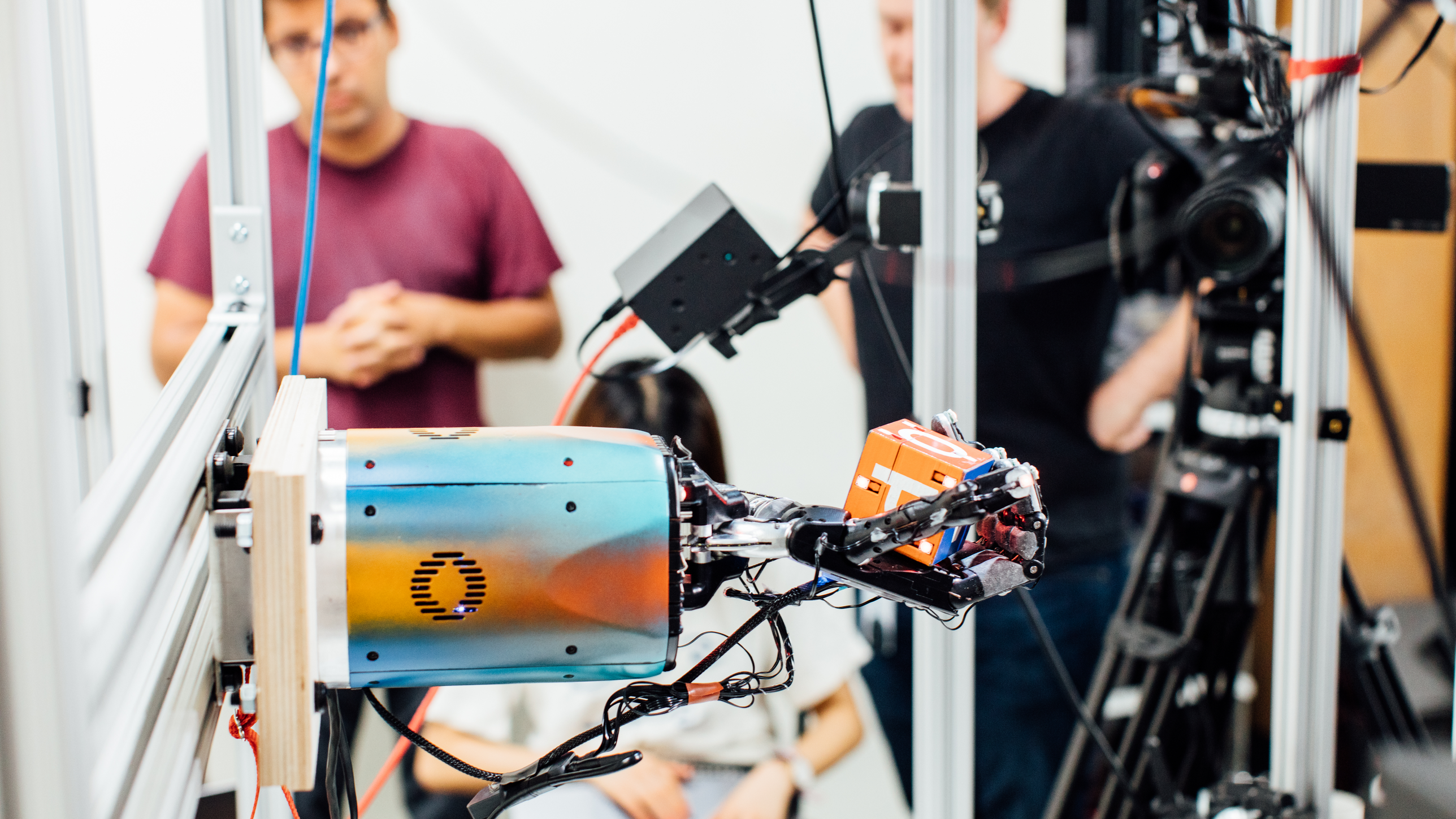


Shadow Dexterous Hand

PhaseSpace tracking




Top RGB camera
Right RGB camera
Left RGB camera

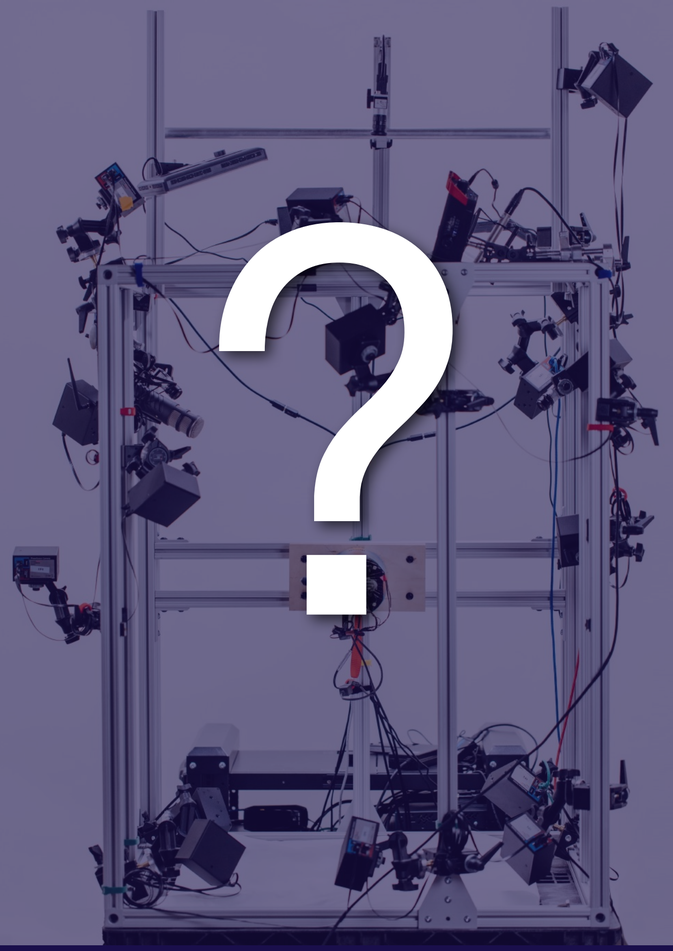


# Sim2Real

SIMULATION ENVIRONMENT

Transfer

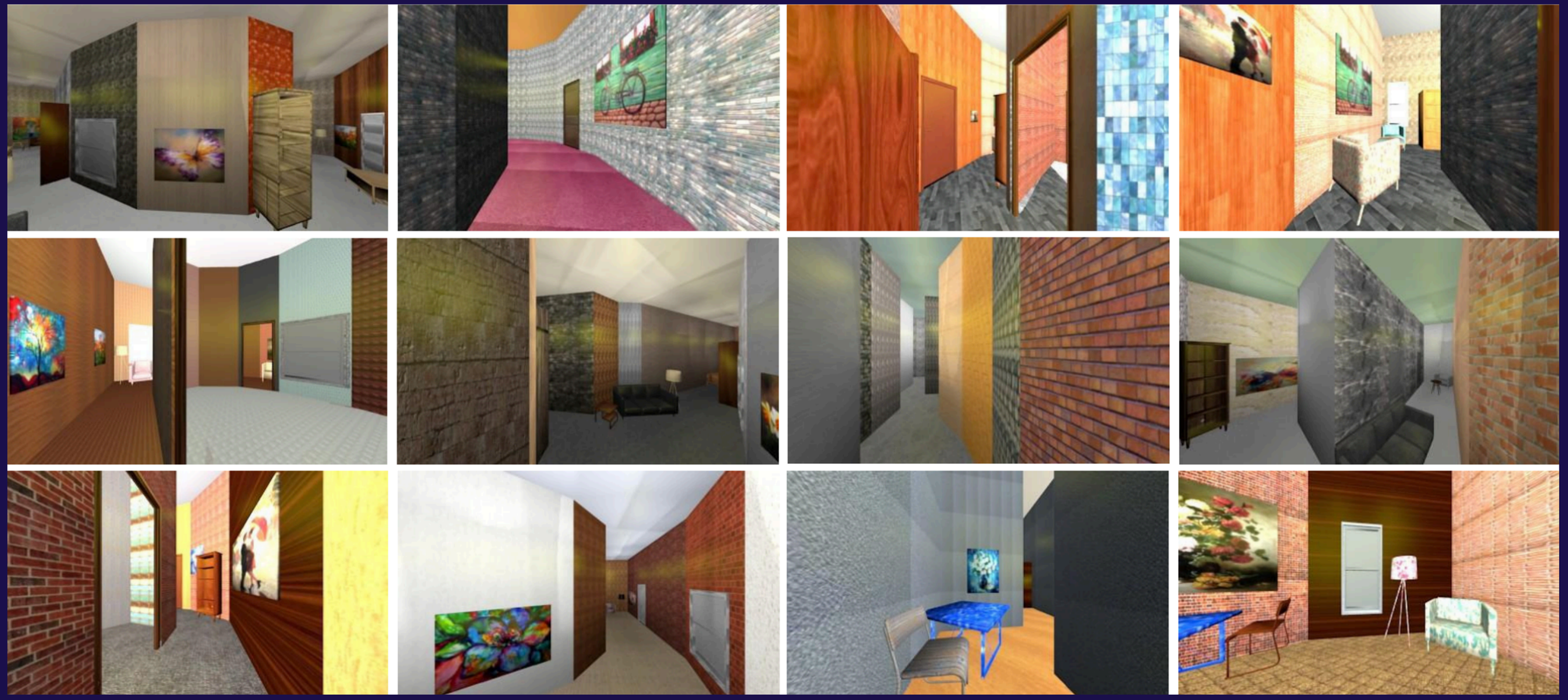
REAL-WORLD ENVIRONMENT



# Reinforcement Learning
# +
# Domain Randomization

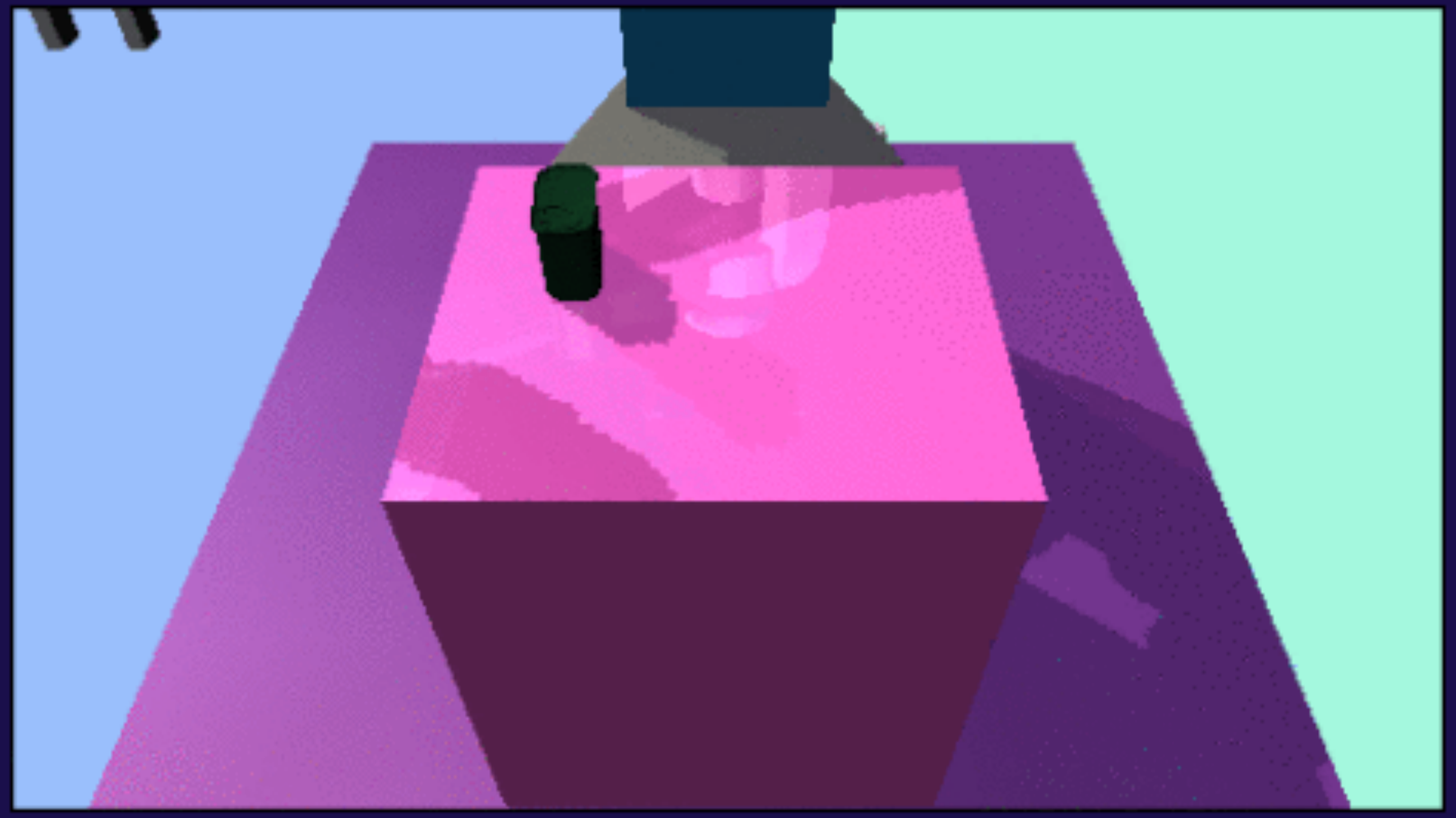
# Domain Randomization

# Domain Randomization



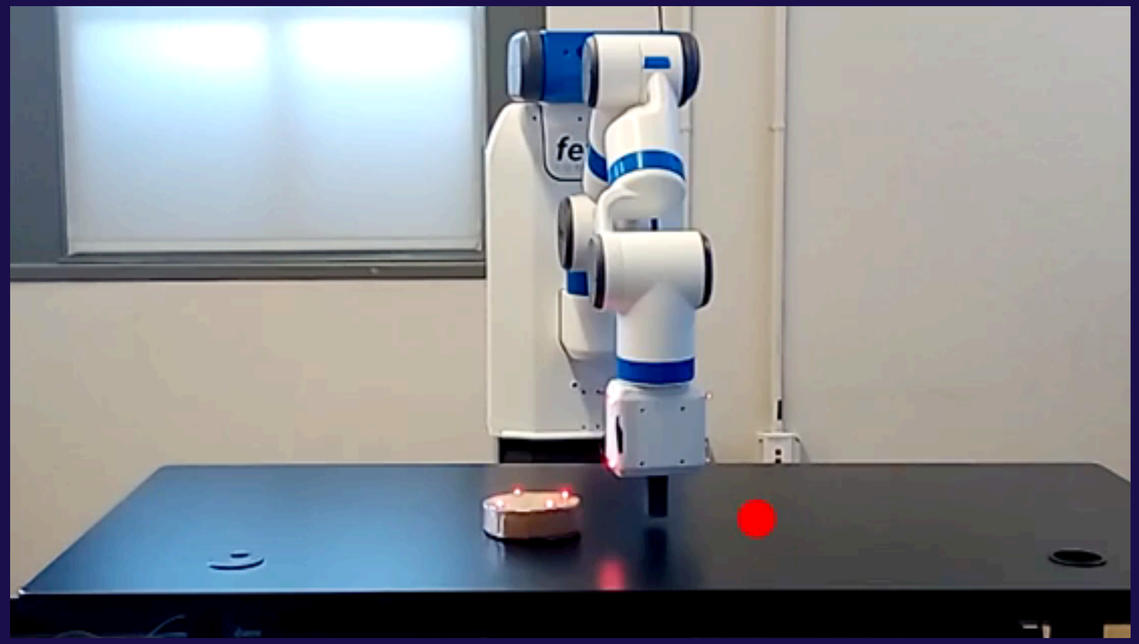
Sadeghi & Levine (2016)

# Domain Randomization

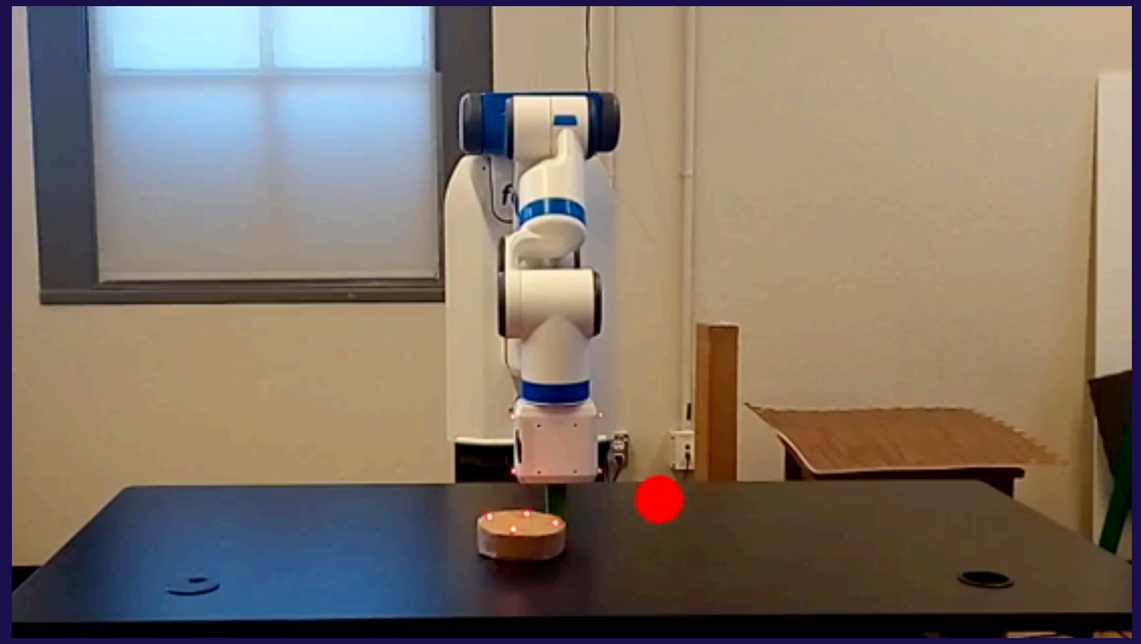


Tobin et al. (2017)

# Physics Randomization



Physics randomization



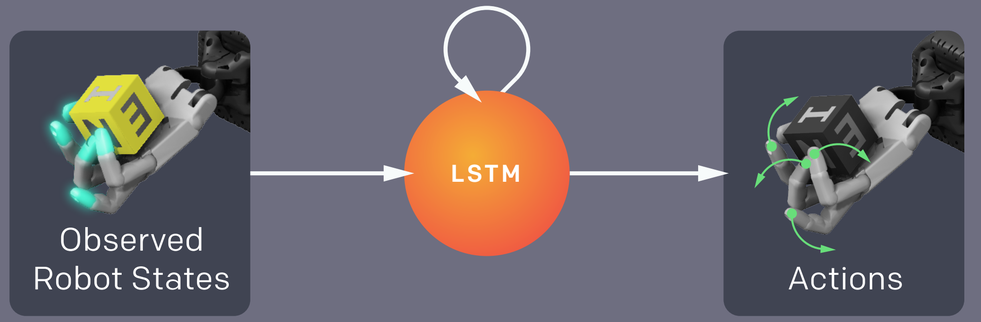
No randomizations

Peng et al. (2017)

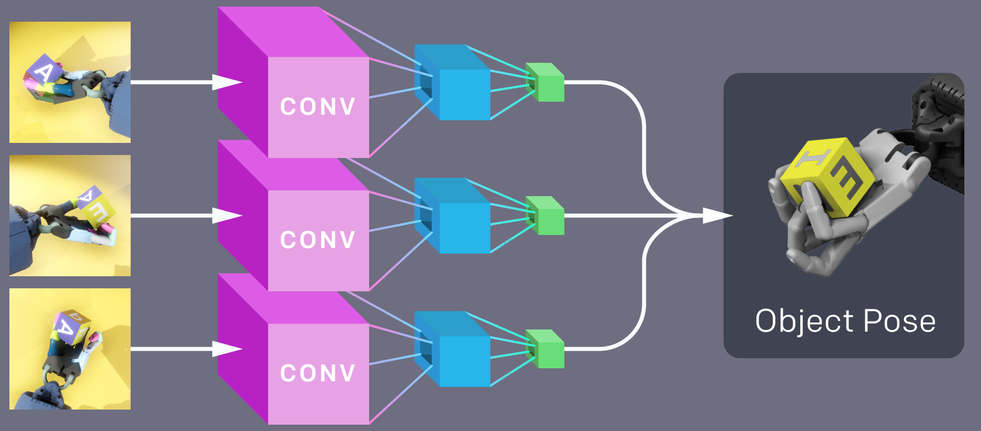
# Our Approach

**A** Distributed workers collect experience on randomized environments at large scale.

**B** We train a control policy using reinforcement learning. It chooses the next action based on fingertip positions and the object pose.



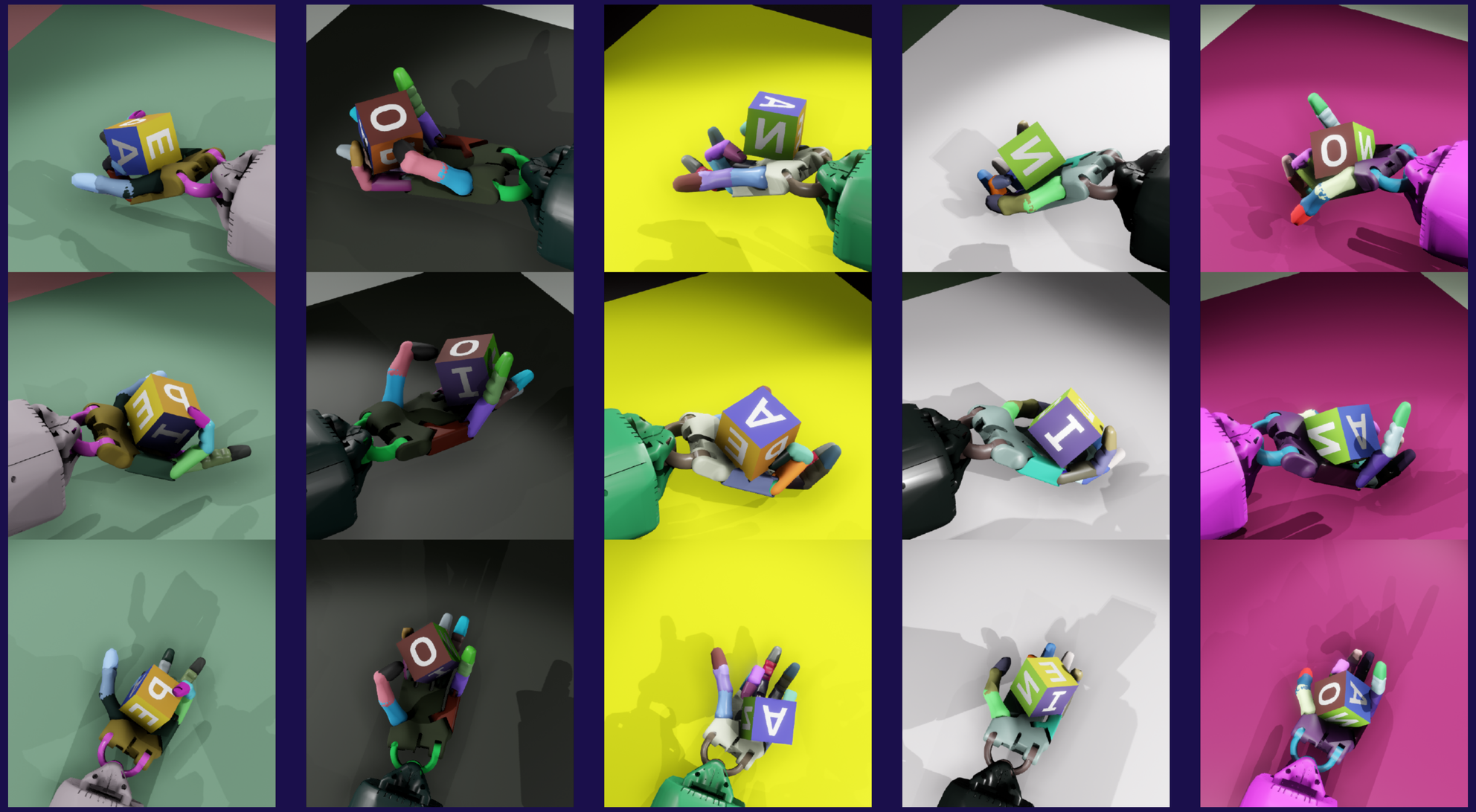

**C** We train a convolutional neural network to predict the object pose given three simulated camera images.



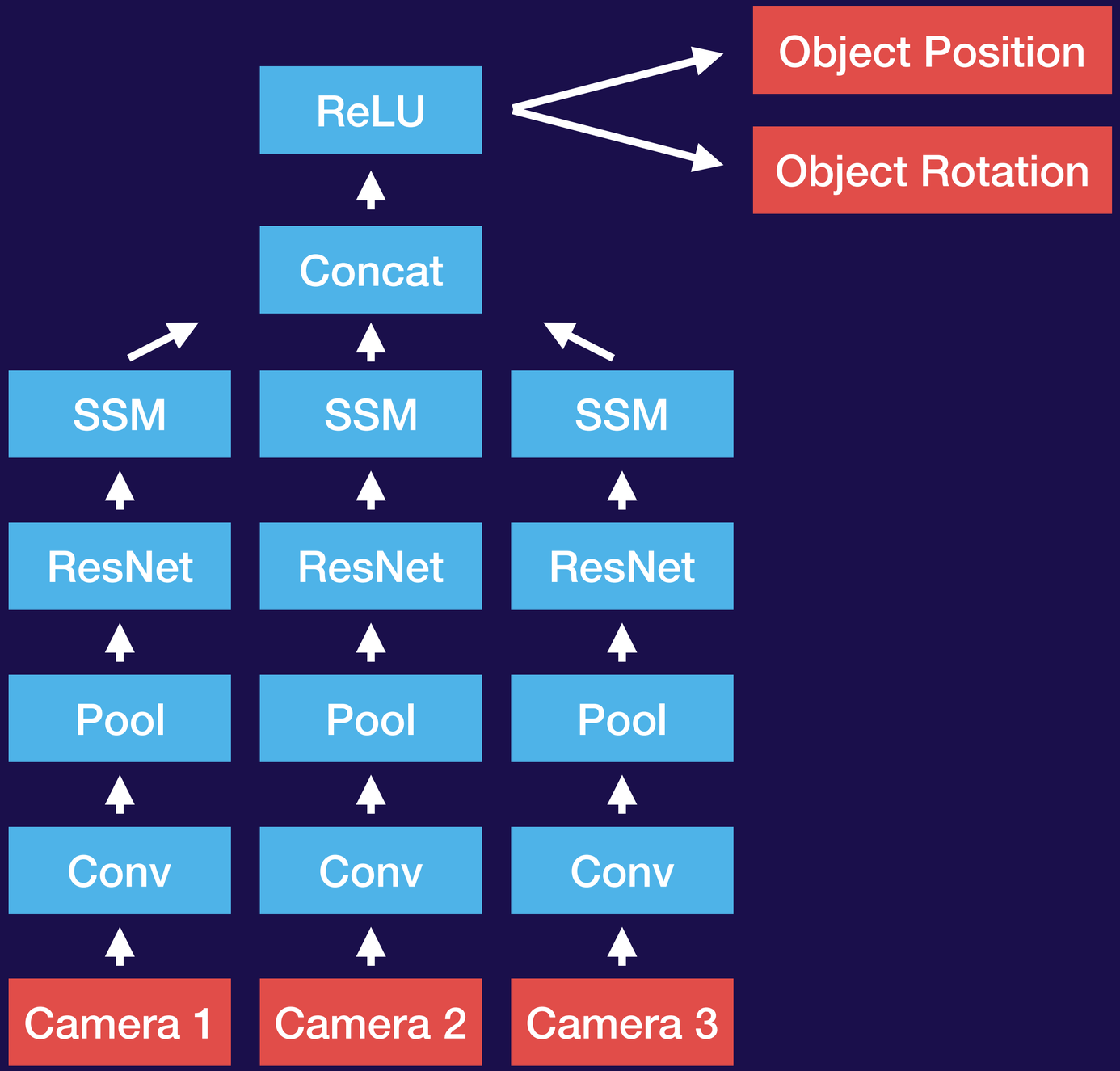
# Transfer to the Real World

**D** We combine the pose estimation network and the control policy to transfer to the real world.

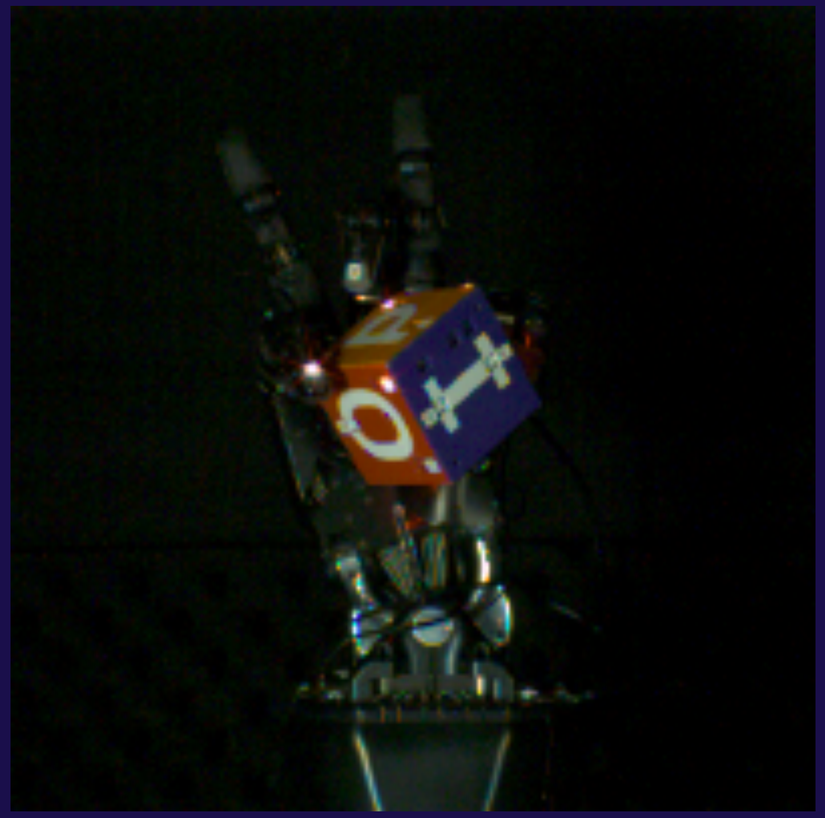
# Appearance Randomizations

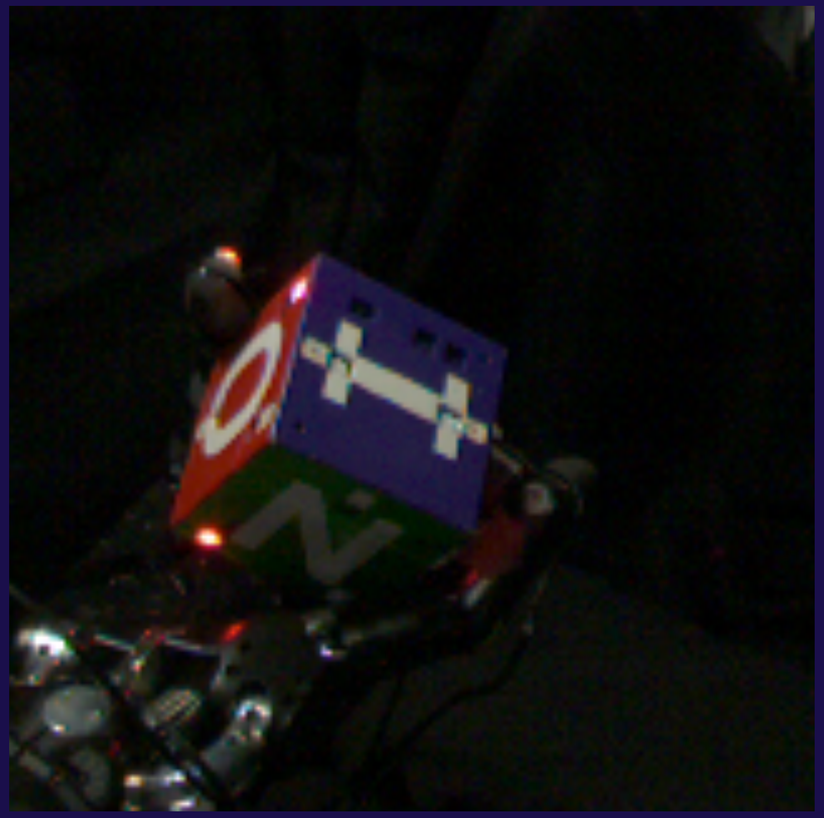


# Vision Architecture



# What the Model Sees





# Physics Randomizations

object dimensions

object and robot link masses

surface friction coefficients

robot joint damping coefficients

actuator force gains

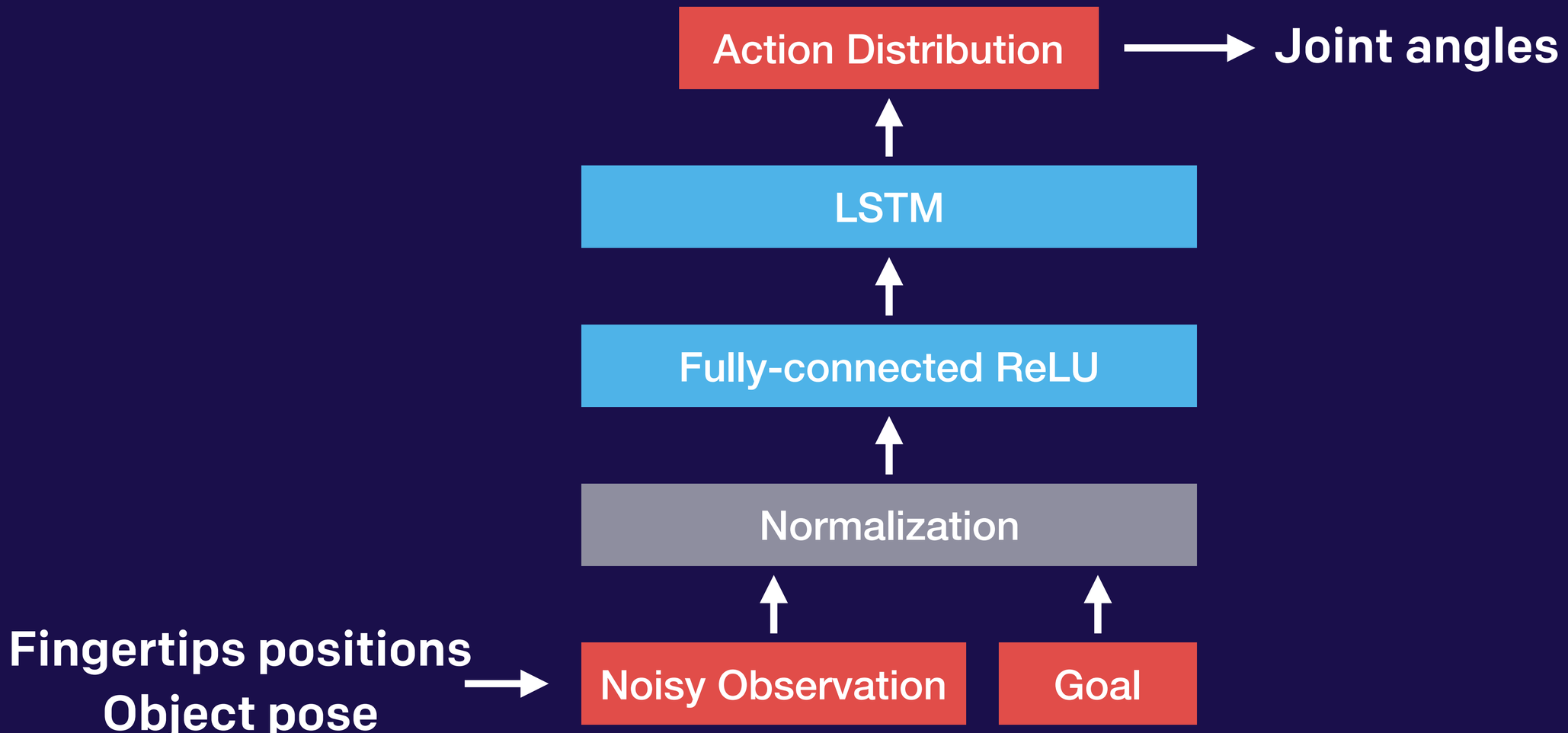
joint limits

gravity vector

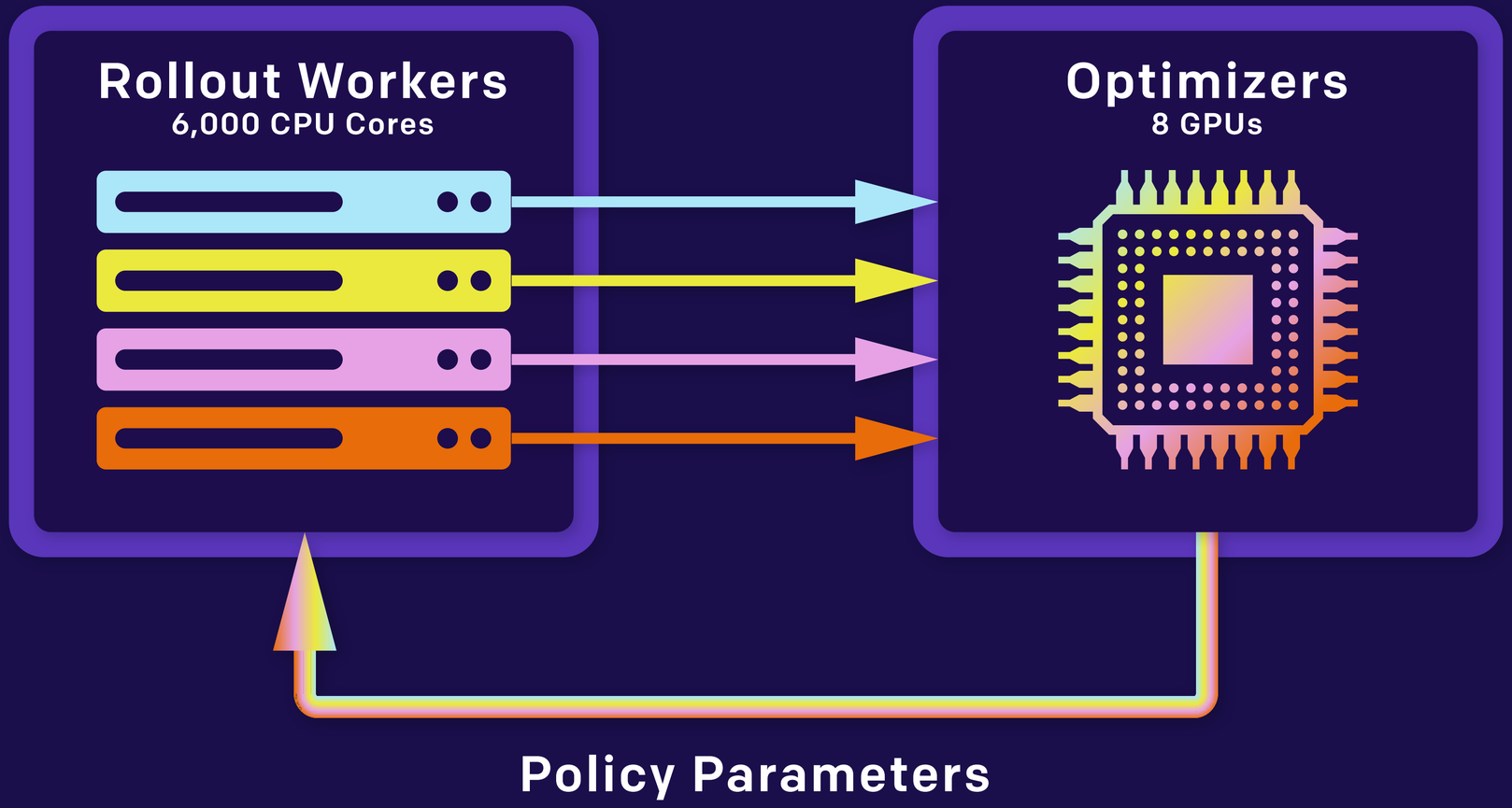
noisy observations

noisy actions

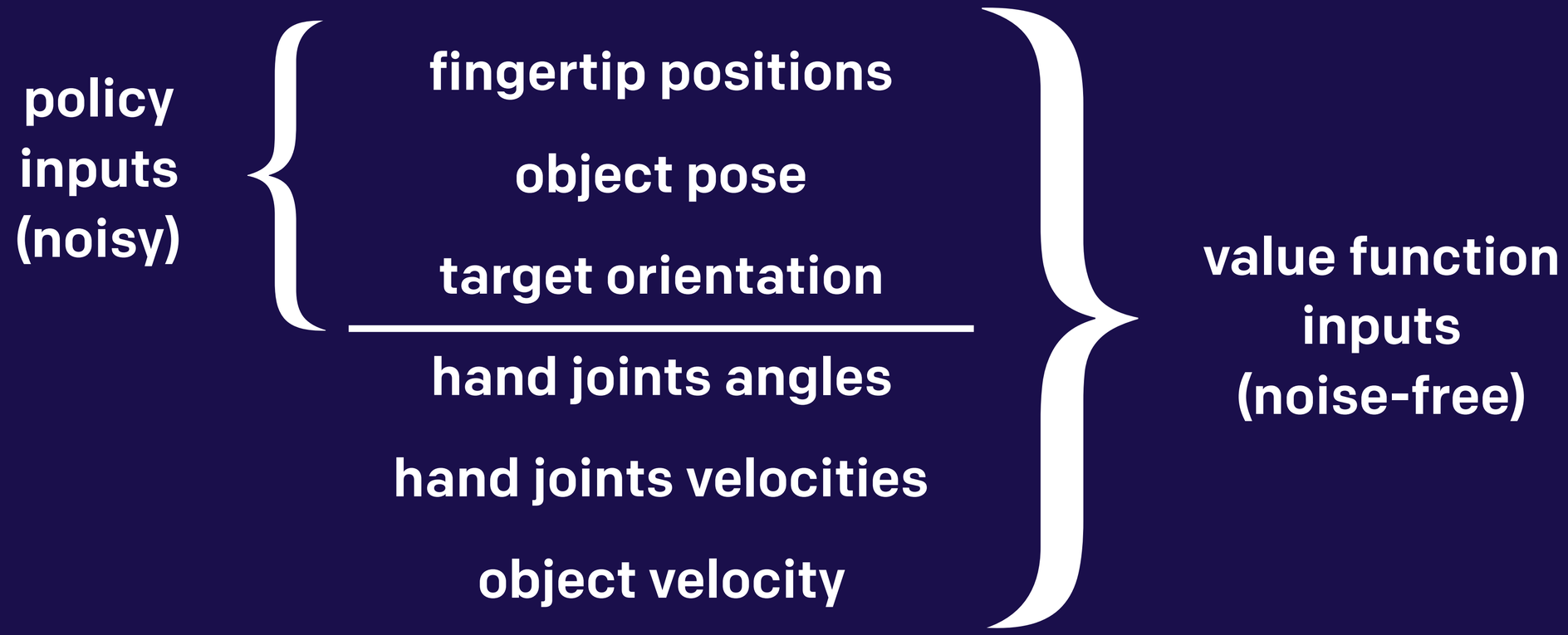
# Policy Architecture



# Distributed Training with PPO

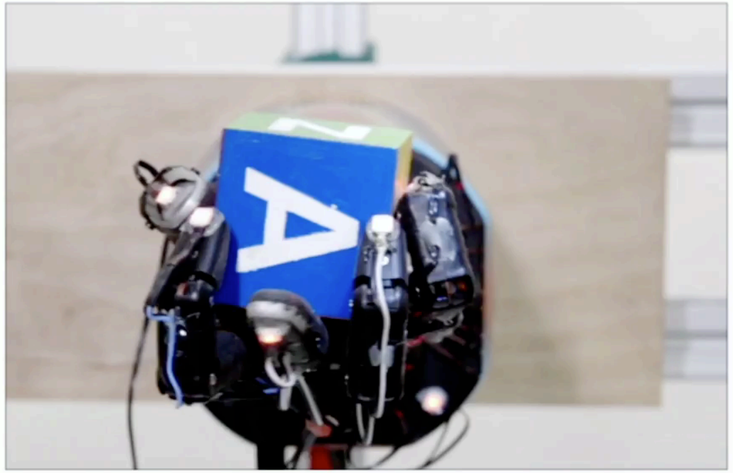


# Value Function

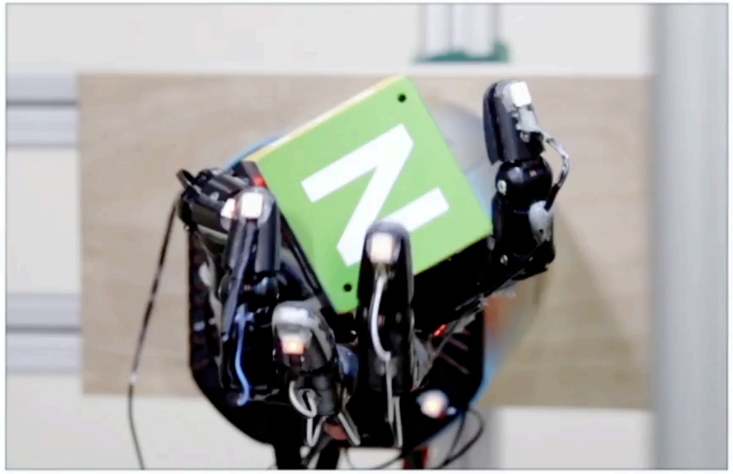


# Results

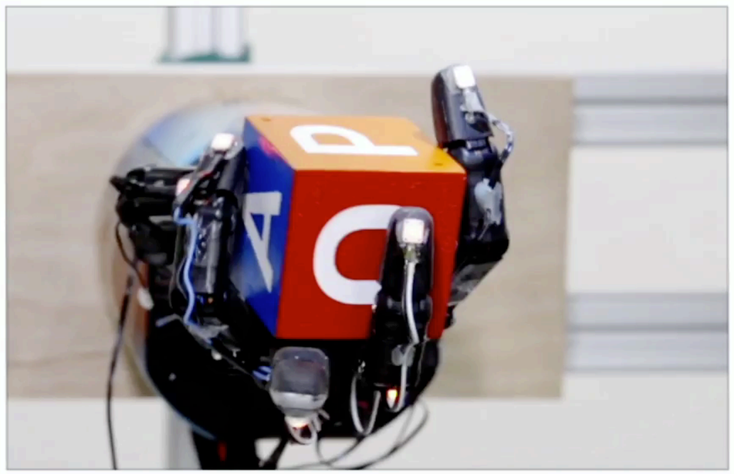
# Emergent Behaviors



FINGER PIVOTING



SLIDING



FINGER GAITING

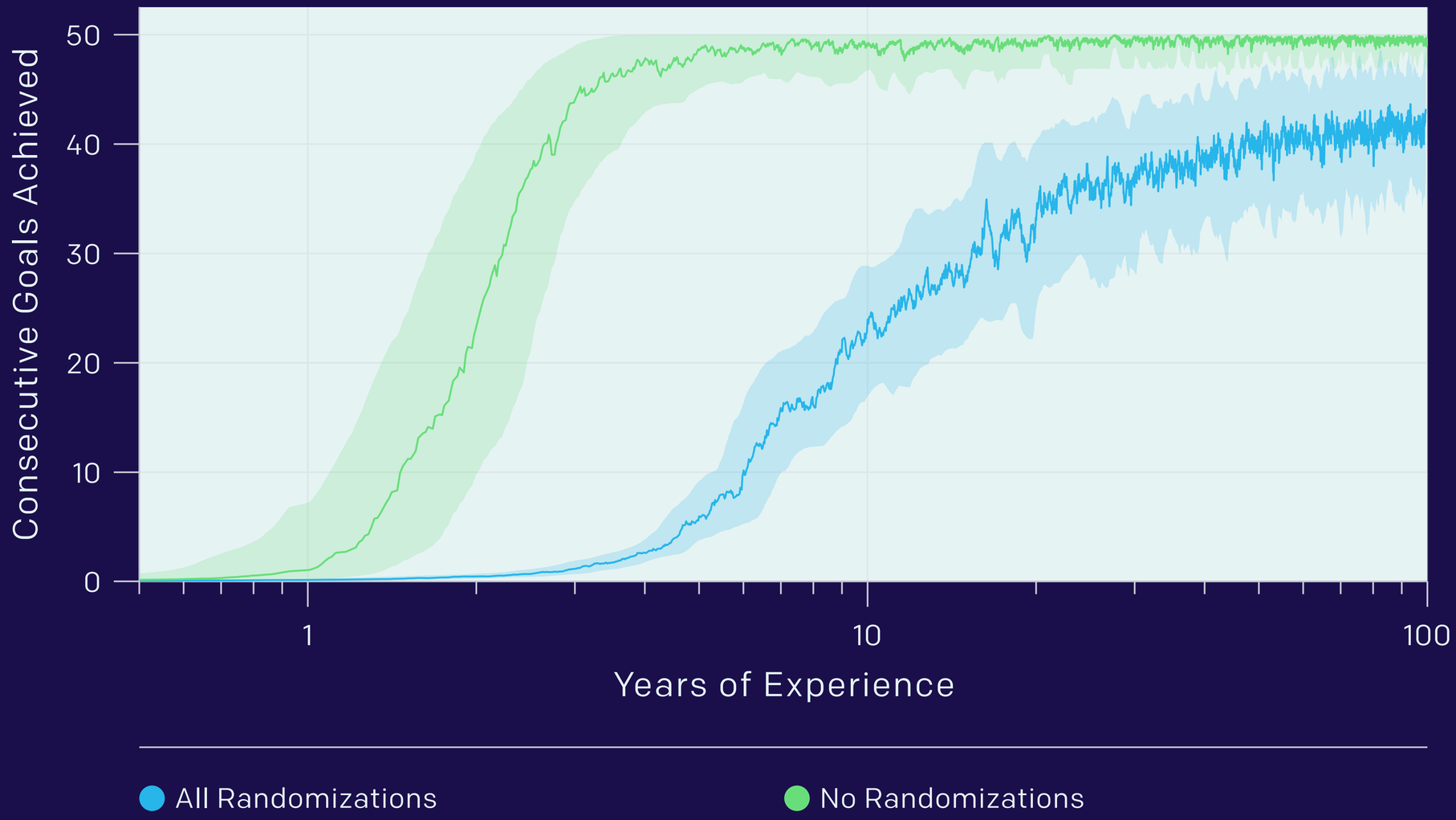
# Quantitative results

| RANDOMIZATONS | OBJECT TRACKING | MEDIAN NUMBER OF SUCCESSES |
|---|---|---|
| **None** | Motion tracking | 0 |
| **All** | Motion tracking | 13 |
| **All** | Vision | 11.5 |

# Effect of Memory

| POLICY | VALUE FUNCTION | MEDIAN NUMBER OF SUCCESSES |
|---|---|---|
| **LSTM** | **LSTM** | 13 |
| **Feedforward** | **LSTM** | 3.5 |
| **Feedforward** | **Feedforward** | 3 |

# Training time



Distribution of environments

+

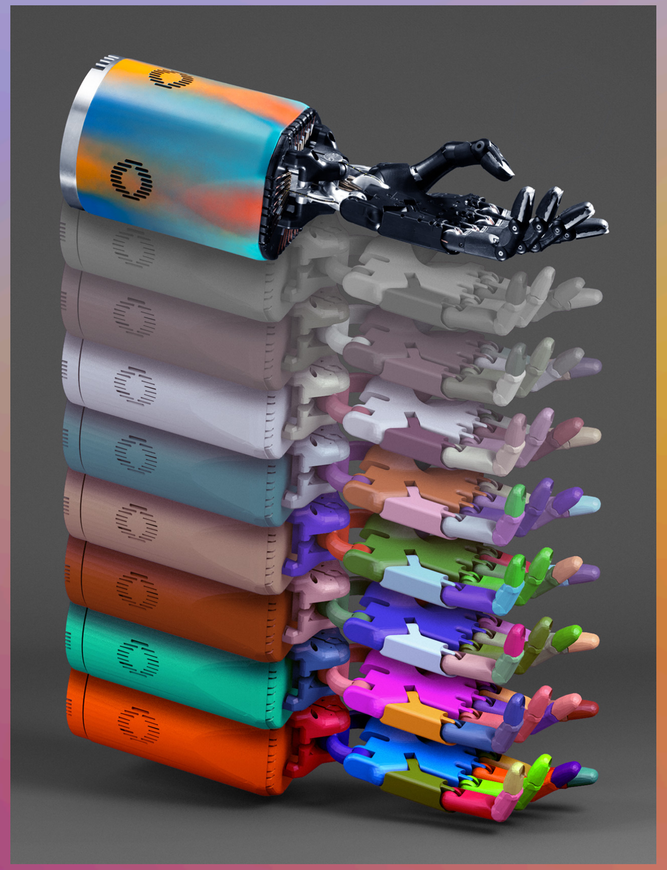
Memory

=

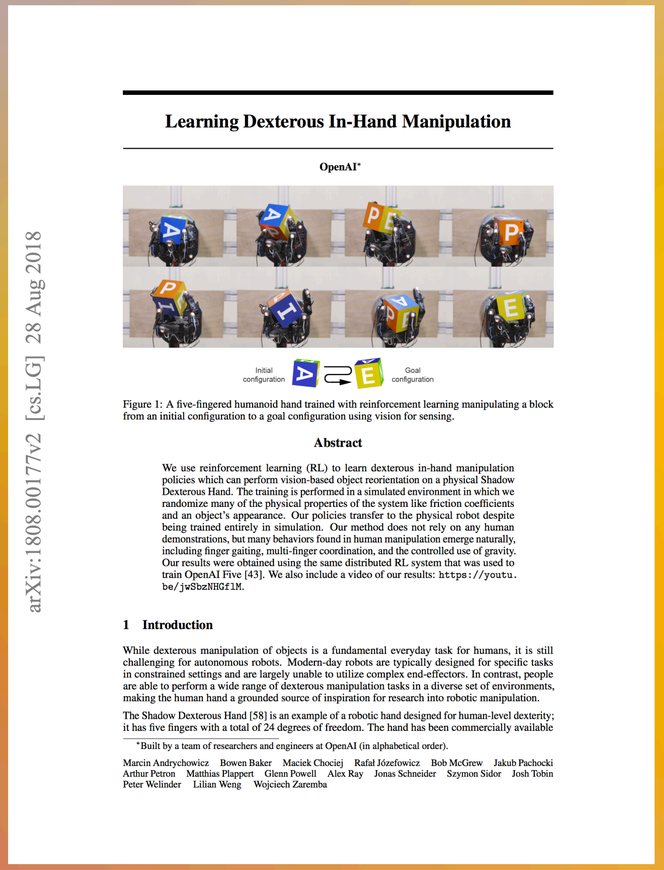
Meta-Learning

# Thank You!

## Blog Post



## Paper



arXiv:1808.00177v2 [cs.LG] 28 Aug 2018

**Learning Dexterous In-Hand Manipulation**

OpenAI*




Figure 1: A five-fingered humanoid hand trained with reinforcement learning manipulating a block from an initial configuration to a goal configuration using vision for sensing.

**Abstract**

We use reinforcement learning (RL) to learn dexterous in-hand manipulation policies which can perform vision-based object reorientation on a physical Shadow Dexterous Hand. The training is performed in a simulated environment in which we randomize many of the physical properties of the system like friction coefficients and an object's appearance. Our policies transfer to the physical robot despite being trained entirely in simulation. Our method does not rely on any human demonstrations, but many behaviors found in human manipulation emerge naturally, including finger gaiting, multi-finger coordination, and the controlled use of gravity. Our results were obtained using the same distributed RL system that was used to train OpenAI Five [43]. We also include a video of our results: `https://youtu.be/jwSbzNHGflM`.

**1 Introduction**

While dexterous manipulation of objects is a fundamental everyday task for humans, it is still challenging for autonomous robots. Modern-day robots are typically designed for specific tasks in constrained settings and are largely unable to utilize complex end-effectors. In contrast, people are able to perform a wide range of dexterous manipulation tasks in a diverse set of environments, making the human hand a grounded source of inspiration for research into robotic manipulation.

The Shadow Dexterous Hand [58] is an example of a robotic hand designed for human-level dexterity; it has five fingers with a total of 24 degrees of freedom. The hand has been commercially available

*Built by a team of researchers and engineers at OpenAI (in alphabetical order).

Marcin Andrychowicz Bowen Baker Maciek Chociej Rafal Józefowicz Bob McGrew Jakub Pachocki Arthur Petron Matthias Plappert Glenn Powell Alex Ray Jonas Schneider Szymon Sidor Josh Tobin Peter Welinder Lilian Weng Wojciech Zaremba

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arxiv.org/abs/1808.00177

| RANDOMIZATONS | OBJECT TRACKING | POLICY | NUMBER OF SUCCESSES | |
|---|---|---|---|---|
| | | | MEDIAN | MAX |
| None | Motion tracking | LSTM | 0 | 6 |
| All | Motion tracking | LSTM | 13 | 50 |
| All | Vision | LSTM | 11.5 | 46 |
| All | Motion tracking | FF | 3.5 | 15 |

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